SMART BIN FOR AUTOMATIC DRY AND WET WASTE SEGREGATION

A DESIGN PROJECT-1 5CS1990 REPORT

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CERTIFICATE

This is to certify that the Design Project -1 work entitled "**SMART BIN FOR AUTOMATIC DRY – WET WASTE SEGREGATION"** is the bonified work done by P.MAHANTHESWARA REDDY (2023BCSE072ED581),B.JASWANTHKUMAR(2023BCSE072ED590),D.DEEPAVALLI(2023BCSE072ED620), P.GNANESH(2023BCSE072ED622) submitted in partial fulfilment of the requirements for the award of the degree Bachelor of Technology in CSE during the year 2025-2026.

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DECLARATION

This is to declare that the report titled "**SMART BIN FOR AUTOMATIC DRY –WET WASTE SEGREGATION"** has been made for the partial fulfilment of the Programme - Bachelor of Technology in **Computer Science and Engineering**, under the guidance of **DR.GOKUL RAJAN V.** I confirm that this report truly represents my work undertaken as a part of **my Design Project -1** work. This work is not a replication of work done previously by any other person. I also confirm that the contents of the report and the views contained therein have been discussed and deliberated with the Mentor.

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ABSTRACT

The growing rate of litter has made environmental concerns increasingly alarming, particularly in densely populated urban areas. Manually sorting dry and wet waste is tedious, unsanitary and unreliable. The Smart Bin for Automated Dry–Wet Waste Segregation has developed a smart-inductive sensor method for identifying and automatically separating dry and wet waste.

The system uses moisture, infrared, and metal sensors, integrated with a microcontroller (Arduino), to register characteristics of waste. When material has high moisture, the waste is identified as wet; when the moisture is low, the waste is identified as dry. The waste will drop into either compartment to be sorted by a servo motor.

As an additional component of the use of the system, the IoT capabilities mean that the bin can provide real time data related to fill level and alerts related to maintenance to a mobile web dashboard, meaning municipal authorities or users can evaluate the time and level for waste collection.

The smart bin is to serve as an affordable, automated, environmentally sustainable, and low-human involvement system with optimal hygienic and sustainable systems for waste management with individuals, organizations, and communities.

LIST OF SYMBOLS & ABBREVIATION

Abbreviation / Component	Description
Arduino UNO	A microcontroller board used to control the sensors and servo motor operations.
Moisture Sensor	Detects the presence of moisture in the waste to determine whether it is wet or dry.
Ultrasonic Sensor	Measures the distance and detects the presence of waste in the bin to automate lid opening or level monitoring.
Servo Motor	Rotates or tilts the waste segregation flap to direct the waste into the respective compartment.

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1. INTRODUCTION

Waste management has become one of the most important problems that the modern cities have due to the exponential population growth, industrialization, and urbanization. The existence of poor waste disposal and segregation can be very disastrous as it may lead to soil and water pollution, undesirable smell, and increased greenhouse gases. Majority of the cities still operate under the old method of waste segregation whereby there are no distinct methods of mixing up the dry and wet wastes before disposing them. These practices identify inefficient recycling since waste is too contaminated to be recycled as well as expose workers to poor and unhygienic environments. The project, thus, aims at providing the Smart Bin under the Automatic Dry-Wet Waste Segregation as one of the first applications of IoT Technology in automatically identifying, evaluating, and classifying waste into two subsets, including dry and wet.

The waste is recognized and characterized as either wet or dry by a Moisture Sensor. An Ultrasonic Sensor is used in sensing the waste to be measured and the Servo Motor is controlled by an Arduino UNO that guides the waste to the correct compartment. Besides, the bin is connected to an IoT platform such as Blynk or ThingSpeak to monitor the level of the bin and the type of waste, which is used to monitor the bin status remotely. The bin has automatic alerts when the bin is full, thereby enhancing the effectiveness of collecting and minimizing overflow. This computerized system reduces human interactions, offers safe and hygienic wastes processing and enhances sustainability through good segregation of wastes. It is inexpensive and can be expanded to any size and it is applicable in homes, businesses or even within the community.

2. LITERATURE REVIEW

Article 1 : Smart Waste Management System based on IoT in Smart Cities (2019).

Abstract: This paper introduces an IoT based system, which assists in real time monitoring of garbage bins. GSM and ultrasonic sensors detect the amount of waste in the bin and alert the authorities: whenever the bin is full. The primary objective was to streamline and streamline the process of waste.

Achievements and shortcomings: The system was able to automate the detection of the waste level and minimization of manual checks. It enhanced time management of waste collection cars. Nevertheless, the system was not able to determine the form of waste (either dry or wet) and the cost of installation was a bit high and thus could not be used at large scale.

Scope: There can be options of waste classification by use of moisture sensors or infrared sensors as well as low-cost hardware in the future to make the system more feasible in both the rural and the city environment.

Paper 2: Arduino based Automatic Waste Segregator (2020).

Abstract: The proposed research is dedicated to one of the simple and low-cost designs that utilize Arduino UNO and sensors to sort waste into dry, wet, and metallic waste. It removes human participation in sorting and promotes cleaner waste disposal activities.

Achievements and shortcomings : The model was effective in identifying and sorting various wastes using simple sensors. It was user-friendly and affordable. Nevertheless, it was manual when it comes to waste input and lacked such features as data monitoring or wireless communication.

Scope: This model can be more sophisticated by adding: the IoT support and automated movement systems and, appropriate to smart city applications.

Paper 3: Implementation of Smart Bin to Separate Waste with a Sensor (2021).

Abstract: This project presented a smart bin with ultrasonic sensor and moisture sensor that detect the classification of the garbage and the fullness of the bin. When the bin starts filling up, a GSM unit will send an alert message to the relevant authority.

Achievements and Weaknesses: The system automated the process of detection and notification fairly well. It employed a mixture of sensors to identify wastes in a better manner. It was however not integrated with mobile applications or cloud platforms and it was only applicable to small scale indoor applications.

Scope: It can be enhanced with IoT dashboard or cloud connectivity in the future to monitor wider. and management.

3. IDENTIFICATION OF THE PROBLEM / SCOPE OF THE PROJECT

Improper segregation of waste is a major environmental and health challenge both in the urban and rural environments. There is a tendency of most of them to confuse dry (non-biodegradable) and wet (biodegradable) waste making recycling and composting more difficult. Such assortment of rubbish not only taxing the sanitation staff but also provides unhealthy environment, contaminating our land and water resources and making it difficult to properly handle the waste. To address this issue, the proposed Dry and Wet Waste Segregation System aims at automating the process of waste classification by sensors and microcontrollers. The system can determine the nature of waste and thus the needy can be sorted into dry and wet bins before being disposed off. The system also has different sensors which include moisture sensors, ultrasonic sensors and IR sensors which measure and classify the waste.

• Levels of moisture are an important indicator—wet waste is indicated by wet moisture whereas dry waste is indicated by dry moisture. The project employs the Arduino-based automation to control the segregation process. Combining this installation in all three places (homes, schools and public areas) would promote improved waste management habits. It reduces the necessity of human involvement, increases hygiene and helps to recycle and compost.

•The project is in line with the smart city projects and enhances sustainable environmental management.

Problem Identified:

• Manual segregation is also uncouth and not only time consuming, but unhygienic too.

• Waste mixture presents a great risk to the environment.

• Knowledge and infrastructure of automated segregation is wanting. The proposed system will also address these challenges since it would provide an economical, automatic, and environmentally friendly solution to waste segregation at the very beginning.

4. PROPOSED IDEA / SOLUTIO

The concept that we are offering here is the development of an automatic mechanism of sorting dry and wet garbage which differentiates and isolates different materials according to the water content and other physical attributes. We aim at providing a low cost, efficient and sanitary method of managing domestic and municipal waste. This system consists of sensors, a microcontroller (Arduino), and a mechanical system that is used to sort garbage into various bins. When garbage is inserted into the system the sensors acknowledge the kind of substance and pass it on to the microcontroller which then instructs the waste to go to the right bin.

The following are the major characteristics of the proposed system:

• Moisture Sensor: It is used to measure the moisture content of the waste. o Wet waste (biodegradable): o High moisture o Dry waste (non-biodegradable):

• Ultrasonic Sensor: This is used to identify the presence and distance of waste, thus automated opening and closing of the bins.

• Arduino Controller: This is the central control unit and it does the sensor data processing and controls the actuator/motor mechanism.

• Servo Motor Mechanism: This sends the waste to the right bin depending on what it is.

How it works:

1. The input area is located close to the waste.
2. The ultrasonic sensor upon detecting the presence of the waste turns on the system.
3. The sensor measures the content of waste to ascertain whether it is wet or dry.
4. Depending on the sensor readings, the Arduino functions and turns the servo motor to get the waste in the right bin.
5. The sorting out of the bins is done automatically as wet and dry and this reduces manual work.

The advantages of the suggested solution:

The effects of sustainable development are that it promotes efficient waste management and maintains a clean environment.

• Decreases human intervention, which reduces the health risks.

• Inexpensive and not complicated to install in homes, office, and community areas.

• Recycles and composts materials by pre- sorting waste. Intelligent city and support sustainability objectives.

5. OBJECTIVES

The primary objective of this design project is to design an automated system that will be able to sort dry and wet waste efficiently with the help of sensors and microcontroller technology. In so doing, we will improve the management of waste at the source, which means they will be properly disposed.

Specific Objectives:

1. To develop and construct an automatic waste separation system using Arduino and the required sensors.

2. To recognize and sort waste in accordance with moisture content, physical characteristics (wet or dry).

3. To minimize human intervention in sorting of wastes, to make the whole process more hygienic.

4.To come up with an effective, affordable and reliable system that can be applicable at home, institutions and in the open. To reduce the workload on the individuals charged with waste management responsibility and increase the efficiency of the municipal waste collection.

5. To be in line with smart waste management systems as Smart City initiatives. In order to design a prototype that is easy to use and maintain and can automatically categorize waste into dry and wet.

Overall Goal: To aid create a cleaner and greener environment by providing a technologically advanced and yet affordable solution to waste segregation at the source.

6. PROCESS / METHODOLOGY

6.1 Identify the needs of customer and users through design thinking methods.

.Through some of the design thinking techniques such as empathizing, defining, and ideation, we managed to identify what customers and users needed in the case of the dry and wet waste segregation system. During the empathize stage, we made observations, interviewed, and surveyed households, sanitation workers, and municipal authorities to have a better understanding of their ordeals with waste management. We found out that sorting waste manually is not only time consuming but also unhygienic and not convenient to do. Many do not know the importance of the separation of wet and dry waste that can lead to mixed waste which is dangerous to the health of workers and causes environmental pollution. The workers of sanitation mentioned that it was necessary to have a system in which they have minimal physical contact with waste, and the residents of households wanted to have a simple and hygienic disposal system.

6.2 Translate the needs of customer into product / process / project.

Based on the consideration of the actual needs of the users, we transformed the concept into an effective design of a product and a process that aims at automating the process of separating wet and dry waste. The principal objective of our customers was to possess a waste management solution that would be easily operated, clean and would need minimum maintenance. This prompted us to develop an automated waste sorting system that will use sensors and microcontroller. This is done by the system by monitoring the moisture content in the waste- high moisture content means wet (biodegradable) waste whereas low moisture content means dry (non-biodegradable) waste. Such measurements are then fed into an Arduino controller, which then directs the waste into the correct bin through a servo motor system.

6.3 Specification of product / process / project.

The Dry and Wet Waste Segregation System was an innovative, automated prototype which was based upon sensors to identify either dry (non-biodegradable) or wet (biodegradable) waste. This system has an Arduino microcontroller and has a moisture sensor, ultrasonic sensor and IR sensor that operate collectively with a servo motor to physically separate the waste. As you insert waste in the input field, the ultrasonic sensor detects its presence, where the moisture sensor detects the amount of moisture to determine the nature of the waste. This information is then fed to the Arduino which manages to get the servo motor moving and sorts the waste out into the right bin.

6.3 Designing the solution for required customer needs.

The Dry and Wet Waste Segregation System transformed to be based on the desires of the customer. mind-oriented-based on automation, hygiene, simplicity, and price-effectiveness. This design technique is

based on design inquiring and person-based engineering, ensuring that it addresses real- efficiently. world waste control issues. Fundamentally, the machine is a totally controlled unit, which is an Arduino. that procedures indicators of various sensors. An ultrasonic sensor is sensing when waste is in position, switching on the device automatically. Then, a moisture sensor analyzes the moisture level of the waste. categorise it as moist or dry. The Arduino microcontroller transmits the information based on this. to a servo-motor, which turns the disposal flap or lid to the ideal bin. This layout

facilitates handlessness, enhancing hygiene and convenience to the users. The components are designed in a small, modular format that allows in to easily maintain and scale, such that it. appropriate to homes, in schools or in a community. The supply is powered by a low-voltage DC supply. machine is power-green and safe to work. Also, it focuses on reliability and

durability, to guarantee extended functionality with least maintenance. By merging technology

and environmental responsibility, this new solution responds to the call to have an automated, dependable and

environmentally friendly waste separation system, which makes the environment cleaner and greener.

6.5 Process / Flow / Design Diagram

Process Description:

1.Waste Detection: The ultrasonic sensor detects the presence of waste at the input section.

2.System Activation: Once waste is detected, the Arduino activates the segregation technique.

3.Moisture Measurement: The moisture sensor analyzes the waste to determine if it's far moist or dry.

4.Decision Making: The Arduino Uno microcontroller strategies the sensor records and decides the waste category.

5.Waste Segregation: The servo motor rotates the flap or lid towards the corresponding bin — one for moist waste and any other for dry waste.

6.Completion: The machine resets and waits for the next waste enter.

Start

↓

Ultrasonic Sensor Detects Waste

↓

Moisture Sensor Measures Moisture Level

↓

Arduino Processes Data

↓

Wet Waste	**Dry Waste**
(High Moisture)	**(Low Moisture)**

↓ ↓

Waste Sent to Wet Waste Bin | **Waste Sent to Dry Waste Bin**

↓

Process End

7 REQUIREMENTS

a.Hardware Requirements

The device uses a number of sensors to bump over and categorize waste automatically:

•Moisture Sensor: Pick up the moisture level in order to be aware of wet or dry trash.

•Ultrasonic Sensor: Determines when and where there is garbage to turn on the device.

• Servo motor: The flap turns to the right bin and motions towards the garbage.

• Arduino Uno: This is the main manage unit, which processes all sensor data and manages.

outputs.

Such sensors cooperate to ensure proper, automated and green waste segregation.

b. Software Requirements

• Operating System: Windows ten / windows eleven.

Consciousness Hardware Programs: Embedded C.

Arduino IDE Development Environment: AdaBoost IDE

|human|>Development Environment Arduino IDE

• Simulation Tool: Tinkercad

technologies CadaCino tone library, Servo library, LiquidCrystal library

of libraries.

• DB / Cloud (where applicable) ThingSpeak / Firebase.

c.Technologies and Tools

Combination of embedded system plays the role of advancing the Dry and Wet Waste Segregation System.

automation with technology and sensor. Arduino microcontroller is the major era that is employed. programming, that is the combination of a large number of sensors to enable automatic identification and segregation of

waste. The system uses IoT concepts in destiny scalability allowing information.

defining or intelligent bin junction in the event of need. Tools that contain the Arduino IDE are utilized in.coding, assembling, and putting the applications on the hardware. Proteus/ Time lapse software or Tinkercad software.program can be used to simulate circuits and prototype. Basic electronic tools like The circuit assembly is done with breadboards, jumper wires and other power substances. Together,those technology and equipment render the system green, low-cost, and easy to implement towaste management applications that are automated.

8 NOVELTY OF THE IDEA / PROJECT

The actual beauty of the Dry and Wet Waste Segregation System is the ingenious and inexpensive design that is simple to operate and makes handling the waste a blessing at the source. In contrast to the outdated manual sorting techniques or costly industrial systems, the automation system in this project involves a smart sensor-based system based on inexpensive components such as the Arduino microcontroller, moisture sensor, ultrasonic sensor, and servo motor. It intelligently recognizes the kind of waste by testing moisture content and classifies it in various containers all by itself thus keeping the environment neat and efficient without requiring human intervention. It has a small and compact modular structure that places it well in the homes, schools, offices, and even in the streets to promote a cleaner and more sustainable lifestyle. This project is a huge step toward an even smarter waste management by integrating simple automation with embedded technology into a simple prototype. Also, it can be scaled as the design can be upgraded later with the features of IoT monitoring or data analytics, which makes it a perfect fit in Smart City projects. Simply put, the originality of the present project is in the simplicity, low cost, and automatization, to provide a feasible and environmentally friendly solution to the challenges of waste sorting in the present day.

9 CONCLUSION

Dry and Wet Waste Segregation System offers an automated, efficient, and hygienic solution to one of the most widespread waste management problems the proper segregation of waste on the origin. With the incorporation of sensors and an Arduino microcontroller, the system will be in a position to detect and classify waste with precision as dry or wet depending on the amount of moisture in it. This automation saves human intervention, lowers the health hazards to the sanitation workers and promotes responsible waste disposal habits.

The project shows that embedded systems and automation technologies can be used to encourage environmental sustainability and facilitate the use of smart waste management programs. It has a low cost, small size and easy-to-use technology making it appropriate to be deployed in homes, institutions and areas. On the whole, the given project helps create a less polluted, more environmentally friendly and sustainable world by easing the waste separation process and encouraging pro-environmental lifestyle.

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PARTIAL IMPLEMENTATION

A partial prototype of the Smart Bin Clicking Automatic Dry-Wet Waste Segregation has been introduced at this stage of development to ensure the functionality of the core part of the system. The partial implementation is based on the fundamental functionality of the sensor integration, information processing with the use of the Arduino, and motor control to perform segregation.

Implemented Components

The setup and testing of the following components and modules have been achieved successfully:

The Arduino UNO: There will be one actuator controller (main microcontroller) which will accept sensor data and pass it over to the actuators.

• Ultrasonic sensor: It is installed to monitor an object that is placed before the bin. The sensor precisely reads the distance and releases the segregation process.

• Moisture Sensor: It is put in place to ascertain the moisture content of the waste. The surpassance of a predetermined amount of reading is considered wet waste, whereas the lower reading is considered dry waste.

Servo Motor: The servo Motor will be linked to the Arduino and programmed to open the waste flap towards an appropriate bin depending on the classification.

Basic Power Supply: The Arduino was powered by a 5V DC power source and connected sensors were powered by the same supply.

Working Demonstration

With the introduction of an object (waste) into the input section:

The sensor is an ultrasonic sensor which identifies the object and transmits a signal to the Arduino.

The moisture sensor is triggered by the Arduino and measures the content of moisture in the waste.

According to the readings, the arduino determines whether the waste is wet or dry.

The lid or the flap is then turned into the corresponding bin by the servo motor to be disposed of.

After the dumping of the waste, the system will restart and wait till the next input.

This shows the fundamental automation cycle of identification → examination four-year-old segregation.

Testing and Results

• The prototype was able to isolate and identify wet and dry waste samples at the initial tests (e.g., paper vs. fruit peel).

• The sensor values were shown in the Arduino IDE Serial Monitor as a debug and verification tool.

• The servo motor mechanism responded to the output of the classification.

Next Steps

• Add IoT connectivity which will be used to transmit bin status to a cloud dashboard.

• Install automatic notification GSM or Wi-Fi.

Design a bigger structure that is enclosed and that can be used practically at home or at the institutions.

• Calibrate sensors using improved accuracy and reliability in mixed wastes.